

Intro to Dat Science - HW 2

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IST 687

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Homework 2

Due Date: 1/26/24

Submission: 01/25/24

```
# Enter your name here: Kent Roller
```

Attribution statement: (choose only one and delete the rest)

```
# 1. I did this homework by myself, with help from the book and the professor.
```

Reminders of things to practice from last week:

Assignment arrow <-

The combine command c()

Descriptive statistics mean() sum() max()

Arithmetic operators + - * /

Boolean operators > < >= <= == !=

This Week: Explore the **quakes** dataset (which is included in R). Copy the **quakes** dataset into a new dataframe (call it **myQuakes**), so that if you need to start over, you can do so easily (by copying quakes into myQuakes again). Summarize the variables in **myQuakes**. Also explore the structure of the dataframe

```
my.quakes<-quakes
str(my.quakes)
```

```
## 'data.frame':  1000 obs. of  5 variables:
## $ lat      : num  -20.4 -20.6 -26 -18 -20.4 ...
## $ long     : num   182 181 184 182 182 ...
## $ depth    : int   562 650 42 626 649 195 82 194 211 622 ...
## $ mag      : num   4.8 4.2 5.4 4.1 4 4 4.8 4.4 4.7 4.3 ...
## $ stations: int   41 15 43 19 11 12 43 15 35 19 ...
```

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
## ✓ dplyr      1.1.3      ✓ readr      2.1.4
## ✓ forcats    1.0.0      ✓ stringr   1.5.0
## ✓ ggplot2    3.4.4      ✓ tibble    3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr     1.3.0
## ✓ purrr      1.0.2
## — Conflicts ————— tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
glimpse(my.quakes)
```

```
## Rows: 1,000
## Columns: 5
## $ lat      <dbl> -20.42, -20.62, -26.00, -17.97, -20.42, -19.68, -11.70, -28.1...
## $ long     <dbl> 181.62, 181.03, 184.10, 181.66, 181.96, 184.31, 166.10, 181.9...
## $ depth    <int> 562, 650, 42, 626, 649, 195, 82, 194, 211, 622, 583, 249, 554...
## $ mag      <dbl> 4.8, 4.2, 5.4, 4.1, 4.0, 4.0, 4.8, 4.4, 4.7, 4.3, 4.4, 4.6, 4...
## $ stations <int> 41, 15, 43, 19, 11, 12, 43, 15, 35, 19, 13, 16, 19, 10, 94, 1...
```

Step 1: Explore the earthquake magnitude variable called **mag**

A. What is the average magnitude? Use `mean()` or `summary()`:

```
mean(my.quakes$mag)
```

```
## [1] 4.6204
```

B. What is the magnitude of the largest earthquake? Use `max()` or `summary()` and save the result in a variable called **maxQuake**:

```
max(my.quakes$mag)
```

```
## [1] 6.4
```

```
max.quake<-max(my.quakes$mag)
```

The magnitude of the largest earthquake is 6.4.

C. What is the magnitude of the smallest earthquake? Use `min()` or `summary()` and save the result in a variable called **minQuake**:

```
min(my.quakes$mag)
```

```
## [1] 4
```

```
min.quake<-min(my.quakes$mag)
```

The magnitude of the smallest Earthquake is 4.

D. Output the **third row** of the dataframe

```
my.quakes[3,]
```

```
##   lat  long depth mag stations  
## 3 -26 184.1   42 5.4        43
```

E. Create a new dataframe, with only the rows where the **magnitude is greater than 4**. How many rows are in that dataframe (use code, do not count by looking at the output)

```
my.quakes.big<-my.quakes %>% filter(mag>4)
num.rows<-my.quakes.big %>% count()
num.rows
```

```
##      n
## 1 954
```

Using the count() function we can see that there are 954 rows.

F. Create a **sorted dataframe** based on magnitude and store it in **quakeSorted1**. Do the sort two different ways, once with arrange() and then with order()

```
quake.sorted1<-my.quakes.big %>% arrange(mag)
quake.sorted2<-my.quakes.big[order(my.quakes.big$mag),]
```

G. What are the latitude and longitude of the quake reported by the largest number of stations?

```
max.stations<-my.quakes %>% slice_max(order_by=stations, n=1)
max.stations
```

```
##      lat  long depth mag stations
## 1 -12.23 167.02   242   6       132
```

```
max.stations2<-my.quakes.big %>% slice_max(order_by=stations, n=1)
max.stations2
```

```
##      lat  long depth mag stations
## 1 -12.23 167.02   242   6       132
```

#Wasn't sure if we were supposed to continue using the modified or original data frame for this one so I used both, the answer is the same regardless since its the max

The latitude for the largest quake is -12.23 and the longitude is 167.02.(12.23 South 167.02 West)

H. What are the latitude and longitude of the quake reported by the smallest number of stations?

```
min.stations<-my.quakes %>% slice_min(order_by=stations, n=1)
min.stations
```

```
##      lat   long depth mag stations
## 1  -21.00 181.66   600 4.4        10
## 2  -23.55 180.80   349 4.0        10
## 3  -16.30 186.00    48 4.5        10
## 4  -20.10 184.40   186 4.2        10
## 5  -15.03 182.29   399 4.1        10
## 6  -19.06 169.01   158 4.4        10
## 7  -17.70 185.00   383 4.0        10
## 8  -21.04 181.20   483 4.2        10
## 9  -27.21 182.43    55 4.6        10
## 10 -18.40 183.40   343 4.1        10
## 11 -20.30 182.30   476 4.5        10
## 12 -14.85 184.87   294 4.1        10
## 13 -17.60 181.50   548 4.1        10
## 14 -20.61 182.44   518 4.2        10
## 15 -25.00 180.00   488 4.5        10
## 16 -17.78 185.33   223 4.1        10
## 17 -20.70 186.30    80 4.0        10
## 18 -21.77 181.00   618 4.1        10
## 19 -21.05 180.90   616 4.3        10
## 20 -17.70 188.10    45 4.2        10
```

```
min.stations2<-my.quakes.big %>% slice_min(order_by=stations, n=1)
min.stations2
```

```
##      lat   long depth mag stations
## 1 -21.00 181.66   600 4.4        10
## 2 -16.30 186.00    48 4.5        10
## 3 -20.10 184.40   186 4.2        10
## 4 -15.03 182.29   399 4.1        10
## 5 -19.06 169.01   158 4.4        10
## 6 -21.04 181.20   483 4.2        10
## 7 -27.21 182.43    55 4.6        10
## 8 -18.40 183.40   343 4.1        10
## 9 -20.30 182.30   476 4.5        10
## 10 -14.85 184.87   294 4.1        10
## 11 -17.60 181.50   548 4.1        10
## 12 -20.61 182.44   518 4.2        10
## 13 -25.00 180.00   488 4.5        10
## 14 -17.78 185.33   223 4.1        10
## 15 -21.77 181.00   618 4.1        10
## 16 -21.05 180.90   616 4.3        10
## 17 -17.70 188.10    45 4.2        10
```

There are multiple observations that have the same number of stations, since it is not known if desired modified code has been included that resolves this issue (if it is an issue)

```
min.stations1.2<-my.quakes %>% slice_min(order_by=stations, n=1, with_ties = FALSE)
min.stations1.2
```

```
##      lat   long depth mag stations
## 1 -21 181.66   600 4.4        10
```

```
min.stations2.2<-my.quakes.big %>% slice_min(order_by=stations,n=1,with_ties = FALSE)
min.stations2.2
```

```
##      lat   long depth mag stations
## 1 -21 181.66   600 4.4        10
```

The latitude of the smallest Earthquake is -21 and the longitude is 181.66 (-21 South 181.66 West) **Step 3:** Using conditional if statements

I. Test if **maxQuake** is greater than 7 (output “yes” or “no”)

Hint: Try modifying the following code in R:

```
if (max.quake > 7) "Yes, Max Quake is larger than 7" else " No, Max Quake is not larger than 7"
```

```
## [1] " No, Max Quake is not larger than 7"
```

J. Following the same logic, test if **minQuake** is less than 3 (output “yes” or “no”):

```
if (min.quake < 3) "Yes, Min Quake is smaller than 3" else "No, Min Quake is not smaller than 3"
```

```
## [1] "No, Min Quake is not smaller than 3"
```